

INTRODUCTION TO DATA MANAGEMENT - 217
PROJECT REPORT
(Project Semester January-April 2025)

Website Analytics Report

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CERTIFICATE

This is to certify that Aayush Garg bearing Registration no. 12311409 has completed INT217. project titled, "Manhole Maintenance Monitoring System" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor:

Ms. Baljeet kaur

Designation of the Supervisor

Asst. Professor

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Date: 13-04-2025

DECLARATION

I, Aayush Garg, student of Introduction To Data Management under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 13-04-2025

Registration No.

12311409

A handwritten signature in black ink that reads "Aayush Garg". The signature is written in a cursive style and is underlined.

Signature

ACKNOWLEDGEMENT

We would like to take this opportunity to express our deep sense of gratitude to all individual who helped us directly or indirectly during this thesis work.

Firstly, we would like to thank my supervisor, Ms. Baljeet Kaur, for being a great mentor and the be adviser I could ever have. His advice, encouragement and the critics are a source of innovative ideas, inspiration and causes behind the successful completion of this project. The confidence shown in me by was the biggest source of inspiration for me. It has been a privilege working with him for the last one year. always helped me during my project and many other aspects related to the program. His talks and lessons only help in project work and other activities of the program but also make me a good and responsible professional.

We would like to take this opportunity to express our sincere gratitude to everyone directly or indirectly involved in the project undertakings of Bank Loan Dashboard Report

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INTRODUCTION

Introduction

In today's competitive digital landscape, data-driven decision-making has become essential for businesses optimize their online presence and improve user engagement. To ensure optimal performance, transparency and strategic planning, it is crucial to have a centralized system that provides real-time insights into webs operations. The Website Analytics Dashboard is designed to serve this purpose by offering a comprehensive visual representation of key metrics such as traffic sources, user behavior, bounce rates, conversion rates, a page performance.

This interactive dashboard enables stakeholders to monitor website trends, evaluate user engagement, ident high-performing pages, and detect potential issues with greater accuracy. By transforming raw data in meaningful visuals, the dashboard enhances operational efficiency and supports evidence-based decision making at all organizational levels. This report provides an analytical overview of the dashboard, its objective key metrics, insights, and strategic implications for improving website performance.

SOURCE OF DATASET

Online sources readily accessible to the public were source of the dataset used in this project in relation movies. Specifically, this dataset has been made obtainable by: Google Dataset Search

Attributes of movies such as title, year release, language, rating, no. of votes, and others are included in dataset. It comprises:

Number of Columns: >10000

Format: CSV(comma-separated values) which was later imported into Excel for preprocessing and the construction of the dashboard.

This dataset is significant for all the analyses in this project and allows different insights to be drawn concerning different metrics of movies over time.

Dataset Preprocessing

The raw dataset contained movie details like release dates, popularity, ratings, genres, etc., but they require several steps for cleaning, transformation, and finally analysis. Below are processes carried out:

1. Removing Null or Missing Values: Rows that contained any null or missing values were removed to uphold the integrity and consistency of the dataset.

2. Renaming of Columns

Columns were renamed (if needed) on a basis of naming convention for consistency and readability:

Overview (though obviously a typo, as encountered in the data; to be corrected in favor of Overview)

Year: Extracted as a four-digit number (e.g. 2021)

3. Creation of Derived Columns

For not available → use average or not available

4. Encoding Issues Repaired

Some of the text fields in the Overview that had encoding errors such as “™”, “”, etc. had those cleaned and replaced with the actual punctuation marks (’, -) via string replace methodology.

Analysis on dataset (for each objective)

Objective 1: Assess the Distribution of Drainage Infrastructure Across Municipalities

The dataset includes drainage infrastructure data, primarily manholes, tagged with their respective **municipality names**. By analyzing the Municipality column in relation to Manhole ID, we can quantify how drainage infrastructure is distributed across different towns or cities in the region.

This analysis helps to:

- Identify municipalities with **the highest concentration of manholes**, which likely require **more frequent inspections and maintenance**.
- Highlight areas with **lower infrastructure density**, potentially indicating **underdeveloped or rural zones**.
- Enable **resource allocation** planning by prioritizing municipalities with larger infrastructure networks.

In the dashboard, a **bar chart** is used to show the number of manholes per municipality, making it easy to compare at a glance. This visualization allows CTDOT or associated authorities to **make data-driven decisions** regarding inspections, budgeting, and infrastructure upgrades.

Key Findings (Example - Replace with real values after charting):

- Hartford had the highest number of manholes, indicating a complex underground network and higher maintenance needs.
- Smaller towns like Essex or Durham had significantly fewer entries, suggesting a more manageable infrastructure scope.

Objective 2: Monitor and Improve Infrastructure Condition

A key field in the dataset is the Condition Rating Inspection Status, which provides insight into the health and functionality of each manhole. Monitoring this parameter is crucial for identifying manholes that are in poor or critical condition, and for prioritizing preventive maintenance.

This analysis helps to:

- Quantify the percentage of infrastructure that has been inspected and rated.
- Identify how many manholes are awaiting inspection, and whether that number is increasing over time.
- Focus attention on assets that might pose a risk of failure, potentially impacting road safety and drainage efficiency.

In the dashboard, a pie chart or bar chart visualizes the condition status distribution, helping stakeholders quickly spot red flags in the system.

Key Findings (Example - Replace with real values after analysis):

- Around 65% of manholes have recorded condition ratings, while the remaining are either marked "Not Inspected" or have missing values.
- 10% of inspected manholes are categorized in "Poor" condition, requiring urgent attention.
- This objective ensures that drainage systems are proactively maintained rather than reactively repaired, supporting CTDOT's long-term sustainability and safety goals.

Objective 3: Optimize Maintenance Operations Based on Ownership and Maintenance Responsibility

The dataset provides valuable details about each manhole's Owned By and Maintained By fields, revealing the

responsible authorities for drainage system upkeep. Understanding these responsibilities is essential for efficient coordination and accountability.

This analysis helps to:

- Identify the distribution of ownership and maintenance responsibilities across different agencies.
- Detect imbalances or overburdened agencies, which may require additional support or reallocation.
- Highlight manholes with unclear or missing ownership data, indicating potential gaps in asset tracking.
- A treemap or stacked bar chart effectively visualizes the share of responsibility among different entities, giving a clear overview of who manages which portions of the infrastructure.
- Key Findings (Example - Replace with real values after visualization):
- CTDOT is responsible for maintaining over 70% of manholes, followed by municipal departments and private agencies.
- Several entries were found with missing or “Unknown” ownership, indicating a need for data validation or field audits.

Understanding this structure allows agencies to plan workload, improve coordination, and ensure no manhole is neglected due to undefined responsibility.

Objective 4: Analyze Historical Installation Trends

The Year Installed field in the dataset allows us to explore how manhole installations have evolved over time. This historical data is crucial for identifying aging infrastructure and estimating the remaining useful life of each asset.

This analysis helps to:

- Track infrastructure development trends over the years.
- Identify vintage manholes that are likely due for replacement or condition review.
- Correlate installation periods with technological upgrades or policy changes in infrastructure planning.
- A line chart or histogram representing the number of installations per year illustrates these trends clearly.
- Key Findings (Example - Replace with real values after analysis):
- A significant number of manholes were installed in the 1990s and early 2000s, indicating a wave of infrastructure growth during that period.
- Very few installations are recorded after 2015, which may suggest data entry gaps or reduced expansion activities.
- This objective supports asset lifecycle management and aids in forecasting future maintenance and replacement needs.

Objective 5: Identify Locations with Traffic Control Needs for Maintenance

The column Requires Traffic Control specifies whether maintenance on a particular manhole will require interruption or control of roadway traffic. This parameter is essential for traffic safety planning, especially on high-traffic highways or urban streets.

This analysis helps to:

- Identify how many assets require traffic control during maintenance.

- Pinpoint specific locations or highways where special attention and coordination with traffic management teams is needed.
- Plan budget and logistics accordingly for such high-effort maintenance tasks.

A bar chart by Maintenance District or Highway filtered to show only those with traffic control requirements provides actionable insights.

Key Findings (Example - Replace with real values after filtering):

- Over 20% of manholes require traffic control, with the majority concentrated in District 1 and along major highways.
- These manholes may incur higher maintenance costs due to traffic management arrangements and should be prioritized for off-peak service scheduling.

This objective directly supports efficient maintenance planning while ensuring minimal disruption to public road usage.

LinkedIn :

https://www.linkedin.com/posts/aayushgarg2306_exceldashboard-dynamicdashboard-datavisualization-activity-7316620375096795136_Dtt?utm_source=social_share_send&utm_medium=member_desktop_web&rcm=ACoAAEdfgtABNnDR7Vrm68aOHhjd2vZv-Ii0u4

Dataset Link : <https://catalog.data.gov/dataset/manholes>



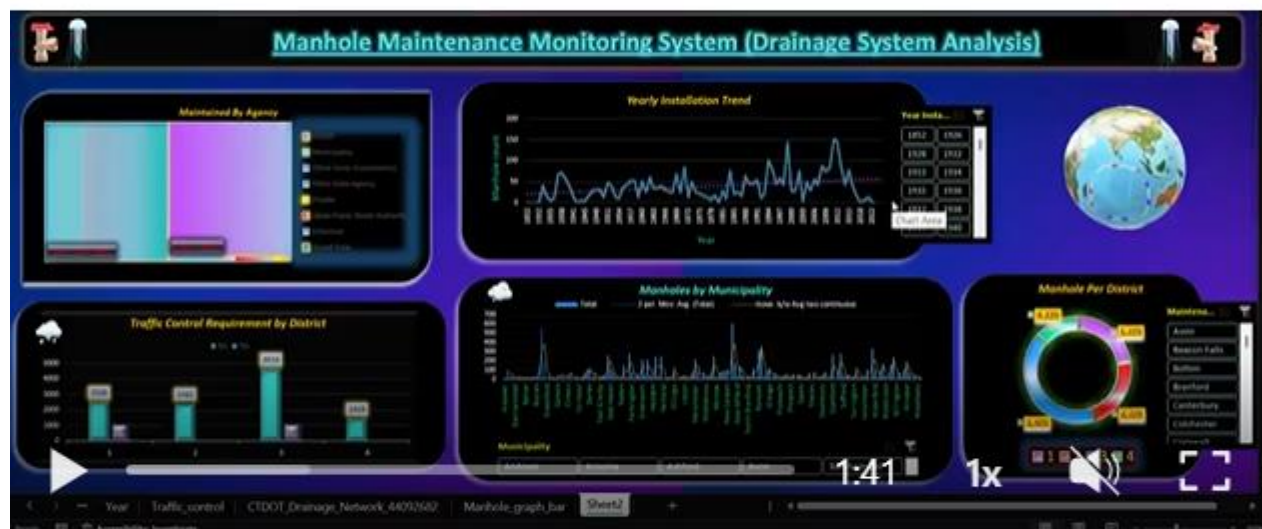
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Innovative Digital Solutions Architect | Turning Vision into Reality | W...

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📊 ✨ From Data Chaos to Dashboard Clarity — My Latest Excel Adventure

This week, I challenged myself to turn a raw, unstructured dataset into something truly insightful and interactive — and that's how my ...more



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SNAPSHOT OF DASHBORD AND PIVOT TABLES

